

KOVARTHINI S

Frontend Engineer | React.js · TypeScript · Real-Time UI · LLM Streaming | 4+ Years

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PROFESSIONAL SUMMARY

Frontend Engineer with 4+ years building scalable web applications and AI-powered products. Twice promoted within 2 years at Fractal Analytics. Contributed to GenAI-powered Smart Search for HDFC and delivered frontend for LLM streaming interfaces, real-time voice products, and multi-client search platforms serving millions of users. Strong track record in performance optimization, real-time UI, and cross-browser delivery in Agile environments.

CORE SKILLS

Frontend: React.js, TypeScript, JavaScript (ES6+), jQuery, Redux Toolkit, Context API, React Hooks, Vite, HTML5, CSS3, SCSS

Real-Time & APIs: WebSocket, LLM Streaming, Streaming API, STT/TTS, REST API, SSO

Performance: Lazy Loading, Code Splitting, Bundle Optimization, PageSpeed Insights, Coverage Analysis

UI Quality: WCAG Accessibility, Responsive Design, Cross-browser Compatibility, BrowserStack

Tooling: AWS S3, GA4, GitHub Copilot, OpenAI Codex, Kilo Code, Git, CI/CD, Agile/Scrum

WORK EXPERIENCE

Engineer Fractal Analytics

Jun 2023 – Present | Bangalore, India

Promoted twice within 2 years: Associate (Apr 2024) → Engineer Gr.9 (Oct 2024) → Engineer Gr.8 (Oct 2025)

- **HDFC GenAI Smart Search:** Contributed to frontend architecture including tech stack selection, project structure, and jQuery to React.js migration — reduced bundle size by ~65% and improved load performance by 40%+
- **Implemented LLM response streaming** on the frontend using Streaming API, rendering real-time AI-generated search results with low latency
- **Applied WCAG a11y compliance and optimization** on HDFC Smart Search; used PageSpeed Insights and coverage analysis to optimize page load performance
- **Axis Bank Smart Search:** Reduced bundle load from ~3s to ~1.3s (55%+ reduction) via lazy loading, code splitting, and race condition fix; handled ongoing bug fixes across HDFC and Axis smart search products
- **Voice Search Assistant:** Built real-time voice transcription interface using WebSocket, STT/TTS, and Streaming API with live AI response rendering and low-latency state updates
- **Delivered 5+ POC chatbot products** for PML Sonic, Reckitt, Philips, Cenomi, and Coherent Global — including voice interfaces, data visualization, and multilingual support
- Resolved critical Safari and iOS-specific rendering bugs across all products; validated via BrowserStack; accelerated development velocity ~20% using GitHub Copilot, OpenAI Codex, and Kilo Code

Project Intern Senseforth (A Fractal Company)

May 2022 – May 2023 | Bangalore, India

Converted to full-time Engineer based on intern performance

- **Optimized HDFC Smart Search on legacy jQuery codebase:** reduced initial load from ~250 KB to ~75 KB (~70% reduction) via code splitting; resolved 40+ bugs on newly developed product
- **Built Tata Capital and Moneyfy Smart Search UI;** resolved critical bugs across Axis Bank and Tata Capital search products
- Developed 10+ reusable React.js components for chatbot and knowledge management products; integrated REST APIs and resolved cross-browser issues via BrowserStack

KEY PROJECTS

HDFC GenAI Smart Search React.js, TypeScript, Redux Toolkit, Vite, LLM Streaming, Streaming API, WCAG

LLM-powered search platform with real-time AI response streaming on the frontend. Led jQuery to React migration, Vite-based performance optimization, a11y compliance.

Voice Search Assistant React, WebSocket, STT/TTS, Streaming API

Live voice transcription dashboard with streaming AI response rendering and real-time state updates. Built for internal POC targeting enterprise clients.

Smart Search Platform React.js, TypeScript, Redux Toolkit, Vite, REST API

Frontend delivery across Axis Bank, Tata Capital, and Moneyfy — performance-optimized, responsive, and cross-browser validated.

Enterprise Chatbot Suite React, Chart.js, MUI, STT/TTS

Multi-client chatbot delivery across PML Sonic, Reckitt, Philips, Cenomi, and Coherent Global with voice interfaces and data visualization.

Knowledge Bot React.js, REST API, Chart.js, Context API, SSO

Enterprise knowledge retrieval interface with SSO authentication, Context API state management, lazy loading for large datasets, and data visualization.

EDUCATION

B.E. – Electronics and Communication Engineering

CGPA: 8.6 / 10.0

Karpagam College of Engineering, Coimbatore | Aug 2019 – Mar 2023

CERTIFICATIONS

Fractal Certified SQL Developer – Associate | Fractal Analytics